

- 1 In this question the number of points on a driver's licence for motoring offences is denoted by  $n$ .

When calculating a driver's premium, a particular insurance company takes account of such offences by multiplying the premium by an amount  $M$ , where

$$M = 1 \quad \text{if } n \leq 3,$$

$$M = 2^{\frac{n}{6}} \quad \text{if } 3 < n < 12.$$

A driver who is insured with this company is paying a premium of £520. Initially he has no points on his licence.

He is convicted of a speeding offence and so receives 3 points on his licence. He is then convicted for dangerous driving and receives a further 6 points on his licence.

Calculate his premium

(A) after his first offence,

(B) after his second offence.

[2]

|      |  |
|------|--|
| 5(A) |  |
|      |  |
|      |  |
|      |  |
|      |  |
| 5(B) |  |
|      |  |
|      |  |
|      |  |
|      |  |
|      |  |
|      |  |
|      |  |
|      |  |

2 In lines 68 to 69, the article says

“Clearly the role of inexperience goes down with age. A possible mathematical model is that it decays exponentially.”

This question investigates this in the case of male drivers. The relevant part of Table 4 is reproduced here.

| Age (in years) | 18.0 | 20.5 | 23.0 | 27.5 | 45.0 |
|----------------|------|------|------|------|------|
| Male %         | 36   | 15   | 6    | 2.5  | 1    |

A model is proposed in which

$$y = a + be^{-k(x-17)},$$

where

- $x$  is the age of the driver,
- $y$  is the percentage of accidents in which inexperience played a role,
- $a$ ,  $b$  and  $k$  are constants to be determined.

(i) Draw a sketch graph of  $y$  against  $x$  for values of  $x$  between 18 and 45.

[1]

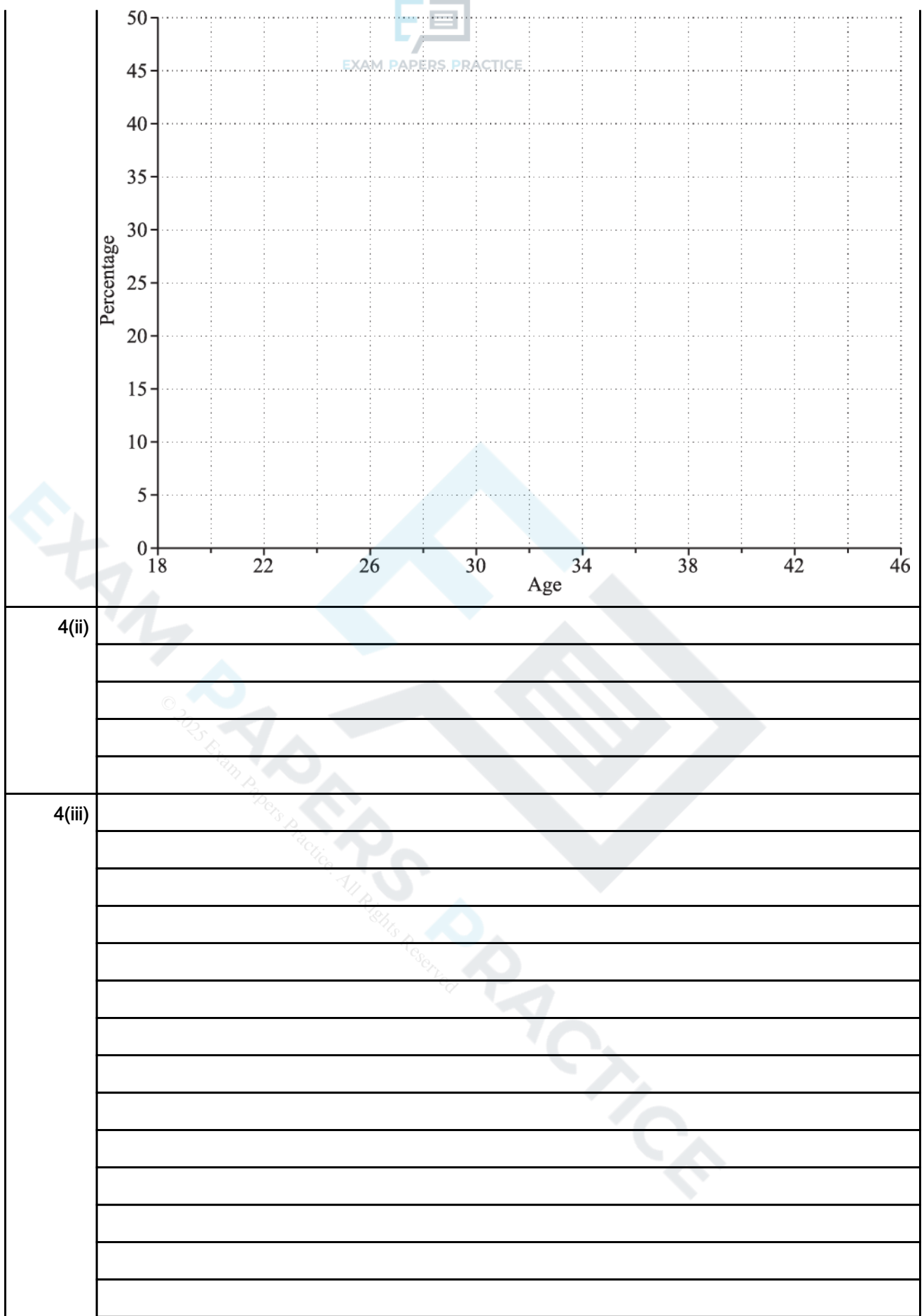
(ii) Explain why the graph justifies putting  $a = 1$ .

[1]

(iii) Taking  $a = 1$  and using the first two data points, it is estimated that  $b = 50.5$  and  $k = 0.3665$ . Investigate whether these values are consistent with the point (23, 6).

[2]

|      |  |
|------|--|
| 4(i) |  |
|------|--|





|  |  |
|--|--|
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |

- 3 On June 1st 2007 Louise paid her first car insurance premium. She did not have any no-claims discount.

She retained her policy with the same insurance company. She had no accidents and so did not make any claim on her insurance. Her no-claims discount followed the pattern in Table 7. Apart from her no-claims discount, her basic premium remained the same.

On May 31st 2014, Louise calculated that so far her no-claims discount had saved her £3875.

What premium did Louise pay on June 1st 2007?

[3]

|   |  |
|---|--|
| 3 |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |



EXAM PAPERS PRACTICE

EXAM PAPERS PRACTICE

© 2025 Exam Papers Practice. All Rights Reserved

4 In lines 97 to 100, the article says

“Most insurance companies have a maximum no-claims discount of 65%. One way of interpreting this practice is that the figure arrived at by applying the maximum no-claims discount is actually the basic cost of the insurance, and that drivers who have not earned the so-called discount are actually paying a surcharge.”

A new driver without any no-claims discount pays  $k$  times “the basic cost of the insurance”.

Find the value of  $k$ .

[2]

|   |  |
|---|--|
| 2 |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |

- 5 The example illustrated in Table 1 in the article referred to the members of a racing club.

This question is about a similar club and the equivalent table is given below. The club has 150 members and each of them belongs to one of the groups P, Q, R or S.

For each group the probability of an accident and the average cost is given in the table below.

| Group                      | P     | Q     | R     | S    |
|----------------------------|-------|-------|-------|------|
| Proportion of people       | 10%   | 20%   | 40%   | 30%  |
| Probability of an accident | 0.1   | 0.15  | 0.05  | 0.2  |
| Average cost per accident  | £5000 | £2000 | £1000 | £500 |

- (i) Complete the table below.

[3]

- (ii) Calculate the premium if each member of the club pays the same amount and the total of the premiums is equal to the total average cost of accidents per year.

[2]

- (iii) The club management decides that the total of the premiums should be 50% more than the total average cost of accidents per year, and that each group should pay an amount proportional to its contribution to the total average cost.

Calculate the premium for a member of group S.

[2]

|      |                                       |       |       |   |   |
|------|---------------------------------------|-------|-------|---|---|
| 1(i) | Group                                 | P     | Q     | R | S |
|      | Number of people                      | 15    |       |   |   |
|      | Average number of accidents in a year |       | 4.5   |   |   |
|      | Average cost of accidents per year    | £7500 | £9000 |   |   |
|      |                                       |       |       |   |   |
|      |                                       |       |       |   |   |
|      |                                       |       |       |   |   |
|      |                                       |       |       |   |   |
|      |                                       |       |       |   |   |

|        |                                                                                 |
|--------|---------------------------------------------------------------------------------|
| 1(ii)  |  |
|        | EXAM PAPERS PRACTICE                                                            |
|        |                                                                                 |
|        |                                                                                 |
|        |                                                                                 |
|        |                                                                                 |
|        |                                                                                 |
|        |                                                                                 |
|        |                                                                                 |
|        |                                                                                 |
|        |                                                                                 |
|        |                                                                                 |
| 1(iii) |                                                                                 |
|        |                                                                                 |
|        |                                                                                 |
|        |                                                                                 |
|        |                                                                                 |
|        |                                                                                 |
|        |                                                                                 |
|        |                                                                                 |
|        |                                                                                 |
|        |                                                                                 |
|        |                                                                                 |
|        |                                                                                 |

6 On lines 94 and 95 it says “Table 4 gives the timings for round trips in which the cars are required to stop at every floor they serve; Table 2 suggests this is a common occurrence in this case”. Explain how Table 2 is used to make this claim.

[3]

|   |  |
|---|--|
| 7 |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |

7 8 people make independent visits to any one of the upper floors of a building with 10 upper floors. What is the probability that at least one of the visitors goes to the top floor?

[2]

|   |  |
|---|--|
| 6 |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |

(i) Referring to Strategy 3 and lines 99 to 101, complete the table below for car C.

[3]

(ii) Calculate the time car C will take to transport all the people who work on floors 7 and 8, and return to the ground floor.

[1]

|      |  |
|------|--|
| 5(i) |  |
|------|--|



## Car C

|                           | Arrival time<br>(seconds) | Departure time<br>(seconds) |
|---------------------------|---------------------------|-----------------------------|
| Ground floor              | 0                         | 20                          |
| Floor 1                   |                           |                             |
| Floor 2                   |                           |                             |
| Floor 3                   |                           |                             |
| Floor 4                   |                           |                             |
| Floor 5                   |                           |                             |
| Floor 6                   |                           |                             |
| Floor 7                   |                           |                             |
| Floor 8                   |                           |                             |
| Floor 9                   |                           |                             |
| Floor 10                  |                           |                             |
| Return to<br>ground floor |                           |                             |

5 (ii)

|  |
|--|
|  |
|  |
|  |
|  |
|  |
|  |
|  |



- 9 In lines 89 and 90 it says "... on average there will be approximately 8 stops per trip. A round trip with 8 stops would take between 188 and 200 seconds". Explain how the figure of 188 seconds has been derived.

[2]

|   |  |
|---|--|
| 4 |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |

- 10 Using the expression on line 51, work out the answer to the question on lines 39 and 40 for the case where there are 10 upper floors and 7 people. Give your answer to 2 decimal places.

[2]

|   |  |
|---|--|
| 3 |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |

11 In line 79 it says “For most journeys, more than half the journey time is composed of load time and transfer time”.  
For what percentage of the journey time for the round trip made by car A in Table 4 is the car stationary?

[2]

|   |  |
|---|--|
| 2 |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |
|   |  |

- 12 In a building with a single lift, the car is located on the ground floor. The following calls arrive in quick succession, all before the car has started on its upward journey.

- Passenger P on floor 9 calls for a down-car (intending to go to floor 1);
- Passenger Q on floor 5 calls for an up-car (intending to go to floor 10);
- Passenger R on floor 4 calls for a down-car (intending to go to floor 1).

The following events need to take place, but not necessarily in the order given.

- E1. Pick up P
- E2. Pick up Q
- E3. Pick up R
- E4. Drop off P and R
- E5. Drop off Q

Assuming the car serves requests as described in lines 16 to 20,

(i) which event occurs first?

[1]

(ii) which event occurs second?

[1]

(iii) which event occurs last?

[1]

|         |  |
|---------|--|
| 1 (i)   |  |
|         |  |
| 1 (ii)  |  |
|         |  |
| 1 (iii) |  |
|         |  |

- 13 Each weekday Alan drives to work. On his journey, he goes over a level crossing. Sometimes he has to wait at the level crossing for a train to pass.

EXAM PAPERS PRACTICE

- $W$  is the event that Alan has to wait at the level crossing.
- $L$  is the event that Alan is late for work.

You are given that  $P(L|W) = 0.4$ ,  $P(W) = 0.07$  and  $P(L \cup W) = 0.08$ .

- (i) Calculate  $P(L \cap W)$ .

[2]

- (ii) Draw a Venn diagram, showing the events  $L$  and  $W$ . Fill in the probability corresponding to each of the four regions of your diagram.

[3]

- (iii) Determine whether the events  $L$  and  $W$  are independent, explaining your method clearly.

[3]

© 2025 Exam Papers Practice. All Rights Reserved

- 14 Jenny has six darts. She throws darts, one at a time, aiming each at the bull's-eye. The probability that she hits the bull's-eye with her first dart is 0.1. For any subsequent throw, the probability of hitting the bull's-eye is 0.2 if the previous dart hit the bull's-eye and 0.05 otherwise.

(i) Illustrate the possible outcomes for her first, second and third darts on a probability tree diagram.

[4]

(ii) Find the probability that

A she hits the bull's-eye with at least one of her first three darts,

[3]

B she hits the bull's-eye with exactly one of her first three darts.

[4]

(iii) Given that she hits the bull's-eye with at least one of her first three darts, find the probability that she hits the bull's-eye with exactly one of them.

[3]

Jenny decides that, if she hits the bull's-eye with any of her first three darts, she will stop after throwing three darts. Otherwise she will throw all six darts.

(iv) Find the probability that she hits the bull's-eye three times in total.

[4]

© 2025 Exam Papers Practice. All Rights Reserved

15 Each weekday, Marta travels to school by bus. Sometimes she arrives late.

EXAM PAPERS PRACTICE

- $L$  is the event that Marta arrives late.
- $R$  is the event that it is raining.

You are given that  $P(L) = 0.15$ ,  $P(R) = 0.22$  and  $P(L | R) = 0.45$ .

(i) Use this information to show that the events  $L$  and  $R$  are not independent.

[1]

(ii) Find  $P(L \cap R)$ .

[2]

(iii) Draw a Venn diagram showing the events  $L$  and  $R$ , and fill in the probability corresponding to each of the four regions of your diagram.

[3]

16 Candidates applying for jobs in a large company take an aptitude test, as a result of which they are either accepted, rejected or retested, with probabilities 0.2, 0.5 and 0.3 respectively. When a candidate is retested for the first time, the three possible outcomes and their probabilities remain the same as for the original test. When a candidate is retested for the second time there are just two possible outcomes, accepted or rejected, with probabilities 0.4 and 0.6 respectively.

(i) Draw a probability tree diagram to illustrate the outcomes.

[3]

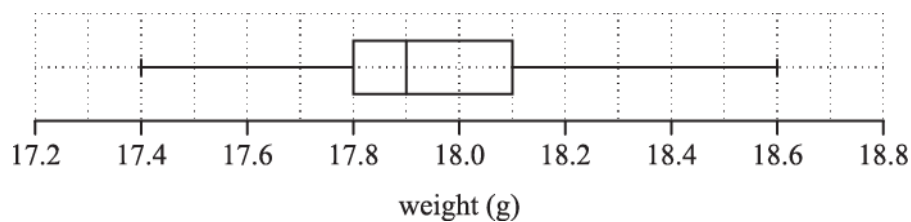
(ii) Find the probability that a randomly selected candidate is accepted.

[2]

(iii) Find the probability that a randomly selected candidate is retested at least once, given that this candidate is accepted.

[3]

- 17 The box and whisker plot below summarises the weights in grams of the 20 chocolates in a box.



- (i) Find the interquartile range of the data and hence determine whether there are any outliers at either end of the distribution.

[5]

Ben buys a box of these chocolates each weekend. The chocolates all look the same on the outside, but 7 of them have orange centres, 6 have cherry centres, 4 have coffee centres and 3 have lemon centres.

One weekend, each of Ben's 3 children eats one of the chocolates, chosen at random.

- (ii) Calculate the probabilities of the following events.

- A all 3 chocolates have orange centres
- B all 3 chocolates have the same centres

[6]

- (iii) Find  $P(A|B)$  and  $P(B|A)$ .

[3]

The following weekend, Ben buys an identical box of chocolates and again each of his 3 children eats one of the chocolates, chosen at random.

- (iv) Find the probability that, on both weekends, the 3 chocolates that they eat all have orange centres.

[2]

- (v) Ben likes all of the chocolates except those with cherry centres. On another weekend he is the first of his family to eat some of the chocolates. Find the probability that he has to select more than 2 chocolates before he finds one that he likes.

[3]

- 18 A normal pack of 52 playing cards contains 4 aces. A card is drawn at random from the pack. It is then replaced and the pack is shuffled, after which another card is drawn at random.

- (i) Find the probability that neither card is an ace.

[2]

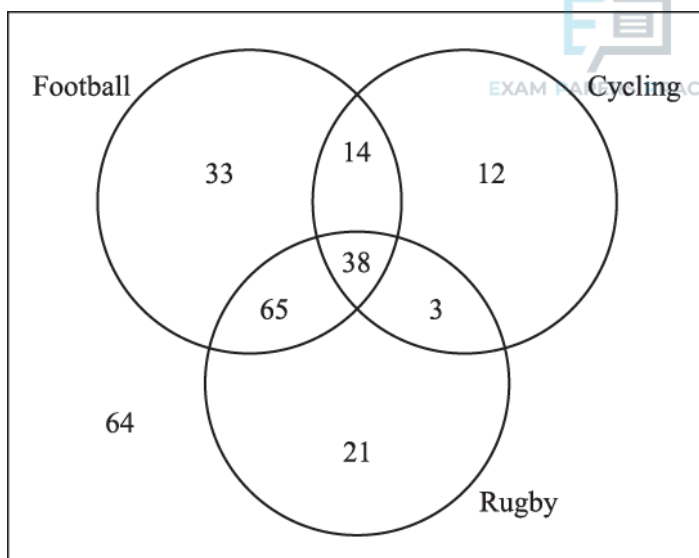
- (ii) This process is repeated 10 times. Find the expected number of times for which neither card is an ace.

[1]

- 19 A survey is being carried out into the sports viewing habits of people in a particular area. As part of the survey, 250 people are asked which of the following sports they have watched on television in the past month.

- Football
- Cycling
- Rugby

The numbers of people who have watched these sports are shown in the Venn diagram.



One of the people is selected at random.

(i) Find the probability that this person has in the past month

A watched cycling but not football,

[1]

B watched either one or two of the three sports.

[2]

(ii) Given that this person has watched cycling, find the probability that this person has not watched football.

[2]

20 Measurements of sunshine and rainfall are made each day at a particular weather station. For a randomly chosen day,

- $R$  is the event that at least 1 mm of rainfall is recorded,
- $S$  is the event that at least 1 hour of sunshine is recorded.

You are given that  $P(R) = 0.28$ ,  $P(S) = 0.87$  and  $P(R \cup S) = 0.94$ .

- (i) Find  $P(R \cap S)$ . [2]
- (ii) Draw a Venn diagram showing the events  $R$  and  $S$ , and fill in the probability corresponding to each of the four regions of your diagram. [3]
- (iii) Find  $P(R | S)$  and state what this probability represents in this context. [3]

21

- (i) There are 5 runners in a race. How many different finishing orders are possible? [You should assume that there are no 'dead heats', where two runners are given the same position.] [1]
- For the remainder of this question you should assume that all finishing orders are equally likely.
- (ii) The runners are denoted by A, B, C, D, E. Find the probability that they either finish in the order ABCDE or in the order EDCBA. [2]
- (iii) Find the probability that the first 3 runners to finish are A, B and C, in that order. [1]
- (iv) Find the probability that the first 3 runners to finish are A, B and C, in any order. [2]

- 22 In a hockey league, each team plays every other team 3 times. The probabilities that Team A wins, draws and loses to Team B are given below.

EXAM PAPERS PRACTICE

- $P(\text{Wins}) = 0.5$
- $P(\text{Draws}) = 0.3$
- $P(\text{Loses}) = 0.2$

The outcomes of the 3 matches are independent.

- (i) Find the probability that Team A does not lose in any of the 3 matches.

[1]

- (ii) Find the probability that Team A either wins all 3 matches or draws all 3 matches or loses all 3 matches.

[2]

- (iii) Find the probability that, in the 3 matches, exactly two of the outcomes, 'Wins', 'Draws' and 'Loses' occur for Team A.

[4]

EXAM PAPERS PRACTICE  
© 2025 Exam Papers Practice. All Rights Reserved

23 Two events  $A$  and  $B$  are such that  $P(A) = 0.6$ ,  $P(B) = 0.5$  and  $P(A \cup B) = 0.85$ . Find  $P(A | B)$ .

[4]

EXAM PAPERS PRACTICE



© 2025 Exam Papers Practice. All Rights Reserved

- 24 There are four human blood groups; these are called O, A, B and AB. Each person has one of these blood groups. The table below shows the distribution of blood groups in a large country.

| Blood group | Proportion of population |
|-------------|--------------------------|
| O           | 49%                      |
| A           | 38%                      |
| B           | 10%                      |
| AB          | 3%                       |

Two people are selected at random from this country.

- (a) Find the probability that at least one of these two people has blood group O. [2]

- (b) Find the probability that each of these two people has a different blood group. [3]

25 Every evening at bedtime my cat Arthur decides whether to spend the night inside or outside the house.

EXAM PAPERS PRACTICE

If Arthur spends a night inside, the probability that he will spend the next night outside is 0.6.

If Arthur spends a night outside, the probability that he will spend the next night inside is 0.9.

Arthur spends a Saturday night inside.

(a) Calculate the probability that Arthur will decide to spend the next Monday night outside. [3]

(b) Show that it is extremely likely that Arthur will spend at least one out of the next seven nights (Sunday to Saturday) inside. [3]

- 26 A particular condition affects 0.8% of the population. 90.1% of the population as a whole carry a certain gene. 9.85% of the population neither carry the gene nor are affected by the condition. Paul discovers that he carries the gene. He believes that it is very likely that he will be affected by the condition. Determine whether or not he is correct. [5]

- 27 Tom is carrying out a survey into the way in which students travel to school. He selects 50 students and asks each of them 'How did you get to school this morning?' The results are given in the table below.

| Walk | Cycle | Bus | Car |
|------|-------|-----|-----|
| 17   | 9     | 13  | 11  |

Tom then randomly selects 4 of these students to interview in more detail.

- (i) Find the number of ways in which Tom can select the 4 students. [2]

- (ii) Find the probability that all 4 of these students walked to school. [2]

- (iii) Find the probability that at least 2 of the 4 students used the same method to get to school. [4]

28 Every day Axel takes the train to work. The probability that he gets a seat on his journey to work is 0.4. The probability that he gets a seat on his journey back home from work is 0.8, independently of whether he gets a seat on his journey to work.

(i) Find the probability that Axel gets a seat on his journey home, but not on his journey to work. [2]

(ii) Find the probability that he gets a seat on at least one of the journeys on a day. [3]

(iii) Given that he gets a seat on at least one of the journeys on a day, find the probability that he gets a seat on both journeys. [3]

29 Sakura and Emily are playing a table tennis match. The winner of the match is the first player to win three games. The probability that Sakura wins a game is 0.55, independently of all other games. Games cannot be drawn.

(i) Find the probability that Sakura wins the match in three games.

[2]

(ii) Find the probability that Emily wins the match.

[5]

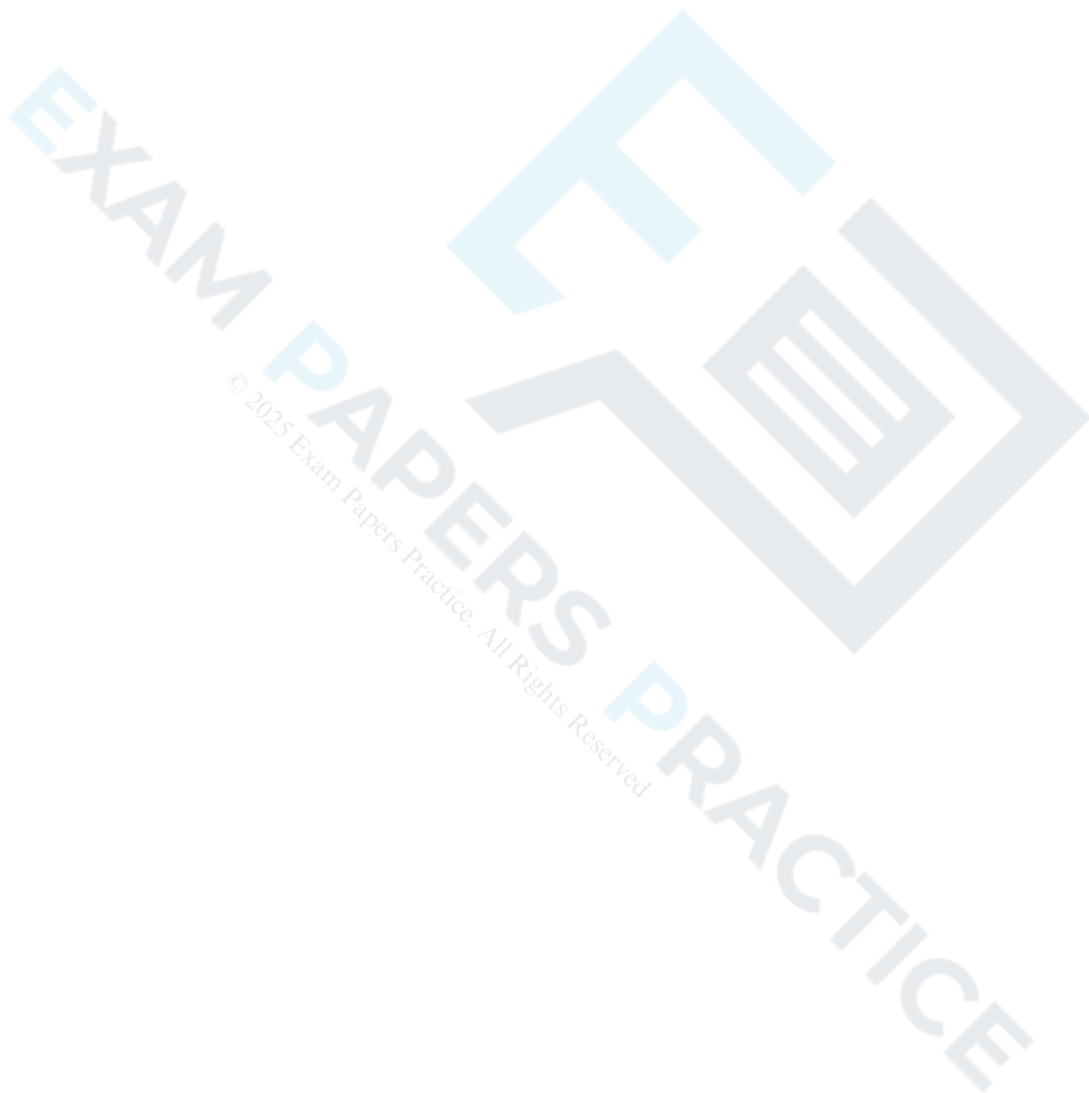
- 30 In a game show, each contestant is asked a number of questions. Each question has a choice of three possible answers.

The probability that John knows the answer to any given question is  $\frac{2}{5}$ .

If John doesn't know the answer he guesses. The probability that he guesses correctly is  $\frac{1}{3}$ .

Calculate the probability that John answers two consecutive random questions correctly.

[4]



31 In this question you must show detailed reasoning.

EXAM PAPERS PRACTICE

The probability that Judith has meat for her evening meal is 0.2 and the probability she has fish is 0.35; otherwise she has a vegetarian dish. If she has meat, the probability that she has indigestion is 0.8, if she has fish the probability that she has indigestion is 0.5 and if she has a vegetarian dish the probability that she has indigestion is 0.1.

Last night Judith had indigestion.

Calculate the probability that she had meat for her evening meal.

[6]

32 The probability distribution of the discrete random variable  $X$  is given in Fig. 4.

|            |     |      |     |     |      |
|------------|-----|------|-----|-----|------|
| $r$        | 0   | 1    | 2   | 3   | 4    |
| $P(X = r)$ | 0.2 | 0.15 | 0.3 | $k$ | 0.25 |

(a) Find the value of  $k$ .

[2]

$X_1$  and  $X_2$  are two independent values of  $X$ .

(b) Find  $P(X_1 + X_2 = 6)$ .

[3]

33 The probability that Chipping FC win a league football match is  $P(W) = 0.4$ .

- (a) Calculate the probability that Chipping FC fail to win each of their next two league football matches. [1]

The probability that Chipping FC lose a league football match is  $P(L) = 0.3$ .

- (b) Explain why  $P(W) + P(L) \neq 1$ .

[1]

In the summer of 2017 in England a large number of candidates sat GCSE examinations in **both** mathematics and English. 56% of these candidates achieved at least level 4 in mathematics and 80% of these candidates achieved at least level 4 in English. 14% of these candidates did not achieve at least level 4 in either mathematics or English.

Determine whether achieving level 4 or above in English and achieving level 4 or above in mathematics were independent events.

[5]



35 In this question you must show detailed reasoning

EXAM PAPERS PRACTICE

A bag contains blue discs and red discs. There are 15 blue discs and an unknown number of red discs. There are more red discs than there are blue discs. A disc is taken at random from the bag and not replaced. A second disc is then taken at random from the bag.

Calculate the probability that 2 blue discs are taken, given that two discs of the same colour are taken. [8]

36  $A$  is the event "Tom forgets to bring his calculator to his mathematics class".

$B$  is the event "Tom forgets to bring his textbook to his mathematics class".

You are given that  $P(A) = 0.5$  and  $P(B) = 0.6$ .

The probability that Tom forgets to bring both his textbook and his calculator to his mathematics class is 0.2.

(a) Calculate  $P(A|B)$ .

[1]

(b) Calculate  $P(A|B')$ .

[3]

(c) State, with a reason, whether or not  $A$  and  $B$  are independent events.

[1]

© 2025 Exam Papers Practice. All Rights Reserved

- 37 Two siblings, Layla and Wilson, have identical smart phones with an app called “Screen Time” which provides the user with statistics concerning the amount of time spent using the screen per day. The statistics for several months of use for Layla’s phone are shown in Fig. 8. Time is measured in minutes.

|                    |      |
|--------------------|------|
| Number of days     | 120  |
| Shortest time      | 12.2 |
| Lower Quartile     | 46.4 |
| Median             | 56.1 |
| Upper Quartile     | 66.1 |
| Mean               | 56.2 |
| Standard deviation | 14.5 |
| Longest time       | 99.8 |

Fig. 8

Layla believes that  $X$ , the amount of screen time she uses per day on her smart phone, may be modelled by the Normal distribution.

- (a) Without doing any calculation, state one feature of the data in Fig. 8 which suggests that Layla may be correct.

[1]

- (b) Verify that the values for the quartiles are consistent with the model

[2]

$$X \sim N(56.2, 14.5^2).$$

- (c) Calculate  $P(X > 90)$ .

[1]

Layla believes that  $Y$ , the amount of screen time used per day by Wilson on his smart phone, may be modelled by  $Y = 2X$ .

(d) Use this model to calculate  $P(Y > 90)$ .

[2]

Each night Layla and Wilson leave their phones on the kitchen table before going to bed. On one occasion their father picked up one of the phones and said:

“This phone has been used for more than an hour and a half today. There is a 99% chance that this is Wilson’s phone.”

(e) Use Layla’s models for the distributions of  $X$  and  $Y$  to determine whether Layla and Wilson’s father’s statement is correct.

[2]

- 38 The 24 members of staff at the Blackley branch of the Midshires Bank decide to start a lottery. They will each pay £5 per week to enter the lottery.

Each week, three of the numbers 1, 2, 3 and 4 will be placed in a random order by a computer. Each member of staff will make 3 different attempts, entered into their computer workstation, to guess what the computer has chosen.

- (a) Show that the probability that a particular member of staff enters the correct numbers in the correct order in any week is  $\frac{1}{8}$ . [3]

If exactly one person chooses the right numbers in the right order, that person will win a prize of all £120 paid by staff to enter the lottery. If **no-one** chooses the right numbers, or if **more than one person** chooses the right numbers, the prize will be 'rolled over' to the next week.

- (b) Calculate the probability that someone will win the prize in the first week. [2]

- (c) Calculate the probability that the customer services manager, Marak, will win the prize in each of the first two weeks.

[2]

In fact, when the lottery started, nobody won until the prize reached £6000. At that point Milena, the chief cashier, won. Marak commented that there was a one in a million chance of this happening to Milena.

- (d) Determine whether Marak's comment is correct.

[4]

39(a) A team called "The Educated Guess" enter a weekly quiz. If they win the quiz in a particular week, the probability that they will win the following week is 0.4, but if they do not win, the probability that they will win the following week is 0.2.

In week 4 The Educated Guess won the quiz.

Calculate the probability that The Educated Guess will win the quiz in week 6.

[3]

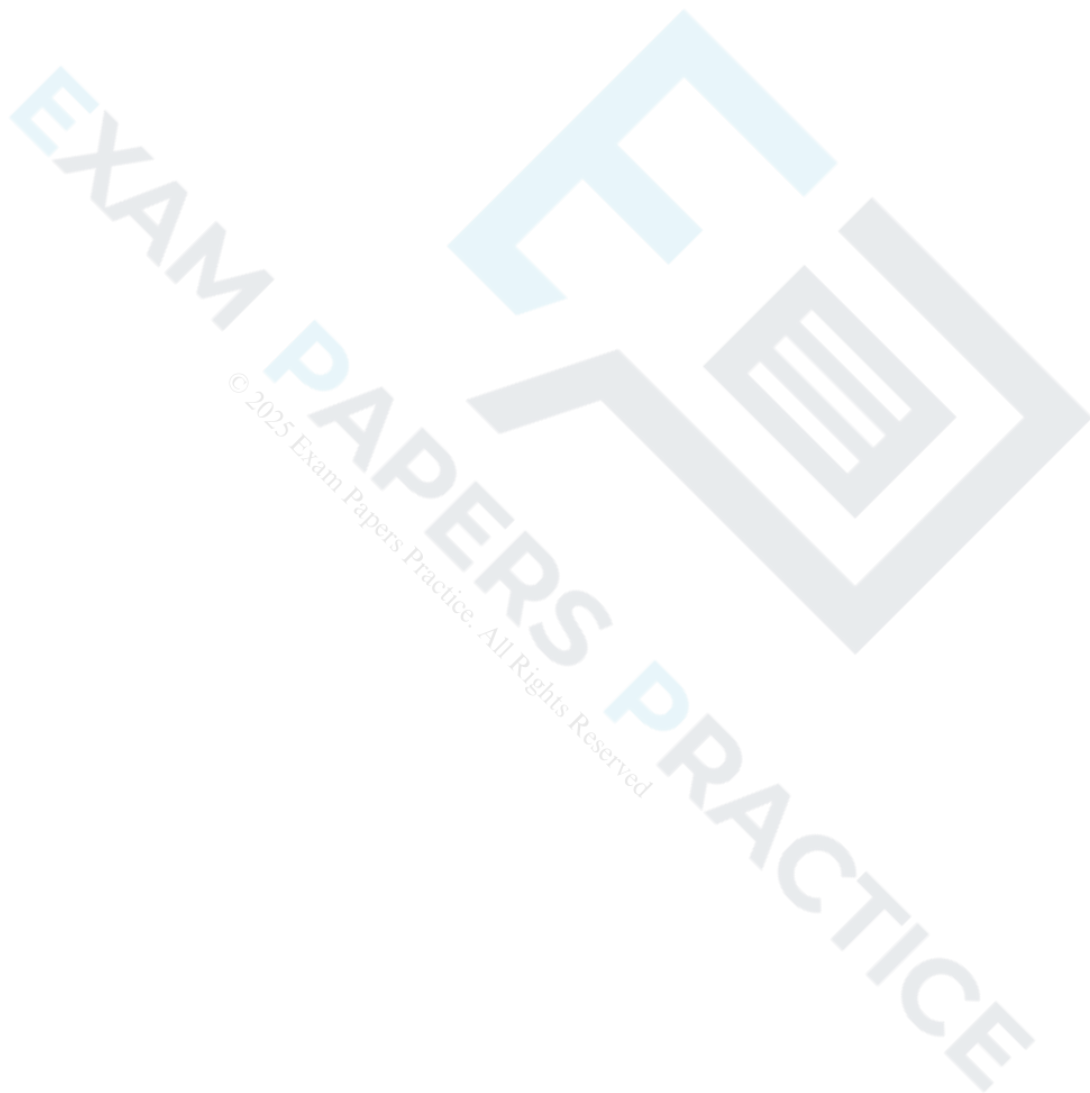
- (b) Every week the same 20 quiz teams, each with 6 members, take part in a quiz. Every member of every team buys a raffle ticket. Five winning tickets are drawn randomly, without replacement. Alf, who is a member of one of the teams, takes part every week.

Calculate the probability that, in a randomly chosen week, Alf wins a raffle prize.

[2]

- (c) Find the smallest number of weeks after which it will be 95% certain that Alf has won at least one raffle prize.

[4]



40(a) Each day John either cycles to work or goes on the bus.

EXAM PAPERS PRACTICE

- If it is raining when John is ready to set off for work, the probability that he cycles to work is 0.4.
- If it is not raining when John is ready to set off for work, the probability that he cycles to work is 0.9.
- The probability that it is raining when he is ready to set off for work is 0.2.

You should assume that days on which it rains occur randomly and independently.

Draw a tree diagram to show the possible outcomes and their associated probabilities.

[3]

(b) Calculate the probability that, on a randomly chosen day, John cycles to work.

[3]

(c) John works 5 days each week.

Calculate the probability that he cycles to work every day in a randomly chosen working week.

[2]



41(a) According to the latest research there are 19.8 million male drivers and 16.2 million female drivers on the roads in the UK.

EXAM PAPERS PRACTICE

A driver in the UK is selected at random. Find the probability that the driver is male.

[1]

(b) Calculate the probability that there are 7 female drivers in a random sample of 25 UK drivers.

[1]



- (c) When driving in a built-up area, Rebecca exceeded the speed limit and was obliged to attend a speed awareness course. Her husband said "It's nearly always male drivers who are speeding." When Rebecca attends the course, she finds that there are 25 drivers, 7 of whom are female. You should assume that the drivers on the speed awareness course constitute a random sample of drivers caught speeding.

**In this question you must show detailed reasoning.**

Conduct a hypothesis test to determine whether there is any evidence at the 5% level to suggest that male drivers are more likely to exceed the speed limit than female drivers. [7]

(d) State a modelling assumption that is necessary in order to conduct the hypothesis test in part (c).

[1]



42(a) A company needs to appoint 3 new assistants. 8 candidates are invited for interview; each candidate has a different surname. The candidates are to be interviewed one after another. The personnel officer randomly selects the order in which the candidates are to be interviewed by drawing their names out of a hat. One of the candidates is called Mr Browne and another is called Mrs Green.

Calculate the probability that Mr Browne is interviewed first and Mrs Green is interviewed last.

[2]

(b) 5 of the 8 candidates invited for interview are women and the other 3 are men. The chief executive can't make up his mind who to appoint so he randomly selects 3 candidates by drawing their names out of a hat.

Determine the probability that more women than men are selected.

[4]

43(a) You are given that  $P(A) = 0.6$ ,  $P(B) = 0.5$  and  $P(A \cup B)' = 0.2$ .

EXAM PAPERS PRACTICE

Find  $P(A \cap B)$ .

[2]

(b) Find  $P(A | B)$ .

[2]

(c) State, with a reason, whether  $A$  and  $B$  are independent.

[1]

44(a) In this question you must show detailed reasoning.

A 5-sided spinner can give scores of 1, 2, 3, 4 or 5. After observing a large number of spins, Elaine models the probability distribution of  $X$ , the score on the spinner, as shown in Fig. 12.

|            |     |     |     |     |     |
|------------|-----|-----|-----|-----|-----|
| $x$        | 1   | 2   | 3   | 4   | 5   |
| $P(X = x)$ | 0.2 | 0.3 | $p$ | $p$ | $q$ |

Fig. 12

When the spinner is spun twice, the probability of obtaining a total score of 9 is 0.06.

Given that  $q < 2p$ , determine the values of  $p$  and  $q$ .

[6]

(b) The spinner is spun 10 times. Calculate the probability that exactly one 5 is obtained.

[2]

- (c) Elaine's teacher believes that the probability that the spinner shows a 1 is greater than 0.2. The spinner is spun 100 times and gives a score of 1 on 28 occasions.

Conduct a hypothesis test at the 5% level to determine whether there is any evidence to suggest that the probability of obtaining a score of 1 is greater than 0.2. [7]

**END OF QUESTION PAPER**